

Aid Allocation with Optimal Monitoring: Theory and Policy

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Abstract

We explore the implications of allowing a poverty-averse donor to monitor aid use within the familiar context of the needs vs. aid effectiveness tradeoff. Although the problem has much broader applicability, we focus on aid allocation between two countries when the donor simultaneously decides about aid shares and country-specific monitoring effort aimed at increasing the amount reaching the poor. Endogenization of aid effectiveness is shown to raise the poor's income in the worse-governed country, yet not necessarily in the better-governed one, whereas the effect on country aid shares is ambiguous. Conventional aid allocation rules should be re-examined in this light. JEL Codes: D02, D86, O22, F35ly

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1 Introduction

Non-interventionist approaches to the allocation of transfer money, in which the idea of controlling its use by the beneficiaries is essentially absent, tend to be increasingly called into question. This is especially evident in the case of development aid: approaches emphasizing policy dialogue, country ownership, and the absence of conditionalities and reform overload, themselves a reaction to the disappointing performances of conditionality-based programmes run by the International Monetary Fund and the World Bank, have led to great disillusionment in the donor community. The institutional weaknesses of many beneficiary countries, and the resulting incidence of aid misuse (Alesina and Weder, 2002), had been clearly underestimated. As recent evidence of massive embezzlement of aid resources, disbursements of aid in a sample comprising the 22 most aid-dependent countries in the world (in terms of WB aid) have been found to coincide, in the same quarter, with significant increases in the value of bank deposits in offshore havens (Andersen et al., 2020). Such correlation suggests that a portion of aid flows is diverted to foreign private bank accounts, most plausibly by members of the local elites (see also Olken, 2006; Easterly, 2007; Platteau, Somville, and Wahhaj, 2014, for more specific evidence of elite capture of foreign aid).

In a context where the use of aid money became increasingly scrutinized by taxpayers in donor countries, the reaction of donor agencies consisted of paying growing attention to governance factors and recognizing the high level of heterogeneity of the developing world. In the wake of this reaction a hard dilemma arose from the observation that the countries evincing the worst governance are also the poorest, that is, aid is least effective where it is most needed (Collier, 2007, Chap. 5; Kanbur and Sumner, 2012). The key notion that logically came to dominate thinking in the development aid debate is that of a trade-off between needs and governance. That notion is encapsulated in the aid allocation rule proposed by Collier and Dollar (2002), or in the Performance-Based Allocation (PBA) formula used by the World Bank's IDA (International Development Association).¹ Because the weight associated to governance indicators in these allocation rules is important, the risk of exclusion for the so-called fragile (the poorest and worst-governed) countries quickly became manifest,

¹See, for instance, World Bank (2017), and note that the formula has not been modified since then.

thereby forcing the concerned aid agencies, the World Bank and the European Commission in particular, to use several types of 'sweeteners' in order to keep these countries eligible.²

A more palatable solution, which is at the core of the present endeavor, consists of acknowledging that aid uses need to be supervised by the donor, and that the intensity of supervision efforts should be inversely related to the quality of governance in the recipient countries. In fact, this is the route increasingly trodden by the donor community, as reflected in the present-day tendency to practice project and programme aid for fragile countries while earmarking budget support and partnership-based aid for better governed and less needy countries. Even within the category of project aid, governance-related constraints can be better overcome through the adoption of more simple project designs, often implying the choice of smaller size and more participative projects (Khwaja, 2009). Not only Western governments and development agencies, but also new donors, such as the United Arab Emirates and China, have come to adopt the new perspective.³

The same trend is also observed for public transfers made inside large entities with heterogeneous membership inside the developed world itself. A striking illustration is provided by the European Union whose poorer member countries are also those where governance is of lower quality. Its leaders have recently agreed to set up a €750 billion fund to help members recover from the COVID-19 crisis. The rule guiding the allocation of this fund is that the countries hardest hit, those of southern Europe, will receive disproportionately large amounts of money even though their governance quality is comparatively low. Revealingly, the Commission has devised a monitoring-cum-sanction mechanism providing that payments may be suspended if the relevant milestones or targets are not fulfilled satisfactorily (Sapir, 2020). Moreover, as attested by the recent withholding of huge EU subsidies (€6.3bn) owed to Hungary (on December 12th, 2022), the initial tranche may not be disbursed if conditions have not been previously established to mitigate the risk of diversion of EU money to elites

²While the PBA formula is based on three unequally weighted indicators -income per capita, a measure of governance, and population size-, the European Commission uses a formula that combines, in a geometric model with equal weights, the same set of indicators, to which measures of human development and vulnerability have been added (Markova, 2013). On sweeteners and the way the PBA rule is being applied, see Guillaumont and Wagner (2015).

³Think, for example, of China's effort to redesign BRI projects by shifting from vast investments to smaller-scale projects (Economist, September 9-15 2023: 49).

close to the government.⁴ The same sort of challenge is facing national governments of developed countries which have to allocate central funds between regions which are unequally prosperous and unequally effective in their use of public money (north and south Italy, for example).

When the way beneficiaries use transfer money is deemed susceptible of improvement, to measure their comparative treatment through aid or transfer shares, as it continues to be done, is potentially misleading. The reason is simply that the donor or transfer-maker has two instruments: aid shares and monitoring intensity. To privilege the first instrument as a guide to the intensity of poverty reduction efforts is unjustified since the two instruments are somehow substitutable. The only meaningful yardstick of aid effort is the outcome of the combined use of these two instruments corresponding to the poverty reduction objective pursued by the donor. More precisely, the correct measure is the post-aid per capita income of the poor, net of residual aid capture by local elites and of monitoring expenses aimed at mitigating such capture. Clearly, these two cost elements (from the viewpoint of the poor) are made interdependent by the optimization of monitoring.

It should be clear now that, although our analytical foray uses the language and framework of aid development literature, it has a much broader scope and relevance. It also bears emphasis that, rather oddly, the question of optimal monitoring has been essentially bypassed in that literature. To address it, we write a two-country model which has three noteworthy features. First, effectiveness of aid use depends not only on the quality of basic or intrinsic governance as it is determined by domestic forces in the recipient country, but also on possible pressures exerted by the donor. This implies that a donor agency uses monitoring and, possibly, punishment mechanisms to obtain an outcome better aligned with its own, poor-oriented preference. Its influence is limited, however, because foreign monitors are unable to improve the basic governance climate, which extends much beyond the domain of donor-funded aid programs or projects. For the same reason, recipient countries may not feel it necessary to modify basic governance within the ambit of aid projects in response to the external discipline imposed by the donor agency. Second, the donor simultaneously chooses the optimal levels for the two available instruments, namely the tailor-made external discipline based on the

⁴Equally interesting is the fact that unlike Hungary, Poland submitted to the conditions imposed by the EU, and henceforth got the green light for access to its share of the fund.

specific needs-governance situation of each beneficiary and the country aid shares (allowing for zero shares). Finally, and unlike in most models found in the literature under concern, the donor's poverty aversion is explicitly featured.⁵

None of the results achieved is trivial and could have been anticipated without the help of a model. A first set of salient results concerns the impact of endogenization of governance - at least the part of governance that affects the use of aid - on aid efficiency, as measured by the portion of aid that eventually reaches the poor after due allowance has been made for residual aid misuse and monitoring expenses. This impact is non-ambiguous: aid efficiency improves in the two countries, yet to a greater extent in the worse-governed country. Variations in country aid shares, however, can move in either direction. Thus, when a donor switches from exogenous monitoring (including zero monitoring) to optimal country-specific monitoring, the aid share of the worse-governed and poorer country can increase or decrease, and the conditions under which it decreases are fully elucidated. This being reckoned, what bears emphasis is the following complementary result: thanks to the donor's comparatively large monitoring effort applied to that country, the post-aid income of its poor is certain to increase, regardless of the way its aid share varies. Such is not necessarily the case for the other, better-governed country where the same switch from exogenous to endogenous monitoring can cause a fall in the aid share that is not fully compensated by a stronger, endogenously determined governance of aid use.

Another original contribution of our approach consists of highlighting the key role of model parameters: the initial level of the gaps between the per capita incomes of the poor and between intrinsic governance quality in the two countries, and the donor's degree of poverty aversion. In particular, these parameters influence the way country aid shares differ when aid allocation under optimal (endogenous) monitoring is compared to the same under exogenous uniform (or even optimal uniform) monitoring. Interestingly, they can work interactively in multiple ways. Regarding poverty aversion, an important result is that strong aversion does not make the donor less willing to seize the opportunity of disciplining fragile countries. What it does, however, is to reduce the donor's willingness to let the aid share allocated to these countries simultaneously fall.

Our effort belongs to standard public economics, as reflected in regional allocation mod-

⁵An exception, here, is Bourguignon and Platteau (2021).

els where a central government has to allocate fixed resources among regions which may use the corresponding amounts more or less efficiently or inclusively. However, a noticeable difference with the optimal redistribution literature, and, in particular, the asymmetric information framework found in the Mirrlees' type of model, is that, in our setting, the central government - here the donor - has full information about the behavior of recipient countries, which are assumed not to change behavior in response to the aid inflow and the monitoring imposed by the donor, except possibly in a very simple and predictive way.⁶ That informational advantage, it must be stressed, does not afford the government full control over the recipients since modifying their behavior involves an increasing cost (disciplining is imperfect). In sum, the simplicity of our own framework regarding informational and behavioral assumptions is the price to pay for endogenizing governance while sticking to the dominant framework used in the aid literature.

The outline of the paper is as follows. Section 2 reviews the directly relevant literature and helps situate the present contribution in its context. Section 3 presents the two-country model with optimal monitoring together with its main properties while Section 4 pursues the formal analysis by comparing exogenous with endogenous donor's discipline. The latter regime corresponds to optimal, country-specific discipline whereas the former stands for either absent external discipline, or uniform discipline set at a level below the one that would be optimal in either country. Section 5 discusses several extensions of the basic model, including the possible response of the recipient countries to the imposition of external discipline, the case of optimal uniform discipline across the two countries, and the case of n recipient countries. Section 6 summarizes the most important results by illustrating them graphically on the basis of numerical simulations. Section 7 concludes and brings out important policy implications of the whole analysis.

2 Literature review

As is apparent from a recent survey (Bourguignon and Gunning, 2017), the literature addressing the issue of aid allocation with a focus on agency problems is surprisingly narrow.

⁶In the public economics literature, a model close to our own setting, but with asymmetric information, is Bordignon et al. (2001).

Azam and Laffont (2003), who provided an earlier survey of aid allocation works using a macroeconomic framework, also proposed a simple and elegant framework for studying this problem. Within a principal-agent framework, the optimal contract between the donor and the recipient country specifies that the latter should receive an aid amount linearly and positively dependent on the poor's level of consumption that it provides.⁷ Such a rule describes in a stylized manner the basic conditionality mechanism whose mode of operation essentially says that aid must be disbursed only after observing the consumption of the poor.

While Azam and Laffont do not explicitly address the issue of aid allocation across different countries, possibly with different internal governance characteristics, Svensson (2000, 2003) tackles this issue upfront in a two-stage game with two recipient countries and a single donor. The two recipients are identical yet subject to independent shocks, so that their ex post situation may differ. The key assumption is that the probability of good states increases monotonically with the amount of reform effort applied by the recipient country, yet this effort is not observed by the donor. In Svensson's model (2000), the second-best contract is a compromise between giving aid to those who most need it and providing optimal incentives for reforms. This translates into the following donor's strategy: in order to induce the recipient to exert higher effort, aid flows in bad states must be lowered and aid flows in good states (more likely to occur when reform effort has been higher) raised (p. 70). Aid thus depends negatively on the extent of (ex-post) poverty so as to preserve effort incentives.

Following the observation that aid effectiveness depends crucially on the quality of institutions and policies in the receiving countries (Burnside and Dollar, 2000), Collier and Dollar (2002) have investigated the problem of a donor's allocation of aid funds between several recipient countries which differ in terms of both policy quality (or governance in our own framework) and poverty. In their setup and unlike in the previous approaches, the quality of policies in each recipient country is taken as given by the donor. The latter then maximizes poverty reduction under that constraint, which leads to the conclusion that, holding the level of poverty constant, aid should increase with quality of policy and, holding policy quality constant, it should increase with poverty. Several papers have broadened or modified the framework proposed by Collier-Dollar, which appears as a key reference in the aid allocation

⁷In an earlier theoretical model of aid conditionality (Adam and O'Connell, 1999), aid was conditional on domestic policy variables, which in turn were determining the welfare of the poor.

literature.⁸

Thus, Carter (2014) also asks how to allocate a given amount of aid between countries with different levels of poverty and governance (see also Carter et al., 2015). Instead of a donor's utility function, however, the author uses the Performance-Based Allocation (PBA) formula of the World Bank's IDA (International Development Association), which features a governance variable and a needs (income) variable, both weighed by a specific parameter. He then identifies the values of these parameters that maximize the discounted sum of the recipients' future utility, assuming that each recipient economy follows a Ramsey growth path under both governance and aid absorptive capacity constraints. Rather intuitively, the presence of the latter constraint implies that, for given governance levels, poorer countries ought not necessarily to receive more aid than richer countries.

Bourguignon and Platteau (2021) also consider governance levels in recipient countries as given. Yet, they provide a more general theoretical framework than the rather specific model used by Collier and Dollar (2002), by explicitly introducing the donor's poverty aversion into the aid allocation problem and by depicting the way the trade-off between needs and governance is played out. This leads them to question the intuitive view that a country which improved its internal governance should always receive more aid from a donor. The validity of such a view turns out to depend on the degree of poverty aversion. In the same paper, the authors consider a variant in which the aid amount received influences aid effectiveness (as measured by its share reaching the poor). As in Carter (2014), the idea is that more aid may worsen the incentives to use it effectively in the recipient country. The conclusions obtained in their basic model continue to hold. Finally, the similarities and differences between the general model offered and the PBA aid allocation formula used by the IDA are brought to light by translating the latter into the language of the former.

In the present contribution, we differ from the aforementioned works to the extent that they are based on the assumption that governance in recipient countries is exogenous or influenced by no other channel than the mere amount of aid transferred (and/or the conditions imposed on the recipient country so that this transfer can take place). Since it allows

⁸See, for example, Cogneau and Naudet (2007) and Wood (2008). Note that the case of several donors and one recipient country is addressed in Knack and Rahman (2004) and Torsvik (2005), who examine how alignment of incentives is affected by the presence of multiple donors. The problem of aid coordination among donors is analyzed in Bourguignon and Platteau (2015).

for donor's (optimal) control of aid use, our endeavor is linked to attempts that have explored how monitoring and sanctioning of individual behavior affect outcomes in other contexts. We think of the fields of law and economics, and the economics of crime in particular (Becker, 1974; Chander and Wilde, 1992; Bose, 1995); the public economics of tax compliance (Andreoni et al., 1998; Kleven et al., 2011) or social allowances (Boone et al., 2007; Setty, 2019); and the economics of (agrarian) contracts (Otsuka et al., 1993). To our knowledge, no application has so far been made to the economics of aid transfers, if we except our own previous work in which the quality of governance is influenced by external discipline in a setup with a single recipient country (Bourguignon and Platteau, 2018).

On the empirical level, two related studies deserve attention because they are rare efforts to assess the effectiveness of monitoring in matters of aid transfers. Based on a review of 1,426 World Bank projects completed between 1981 and 1991, the first study highlights the potential contribution of monitoring to effectiveness of project aid (Kilby, 2000). It leads to the following conclusions: (i) past supervision has a positive and perceptible impact on project performance; (ii) early supervision is much more effective than later supervision; (iii) supervision is profitable; and (iv) the impact of supervision is relatively homogenous across regions, sectors and macroeconomic conditions.⁹ Using the same dataset, the second study, by Chauvet et al. (2017), argues that not only a more precise supervision of projects increases the likelihood of project success, but the effect of higher monitoring precision is significantly more effective when interests between donor and recipient (as perceived by the donor) are more diverging.

3 The model and its properties

3.1 The setup of the model

Before presenting the main ingredients of the model, a few remarks are in order. First, the perspective adopted is clearly normative, and it justifies the (convenient) assumption of a single donor in charge of allocating the available aid. Any coordination failure between mul-

⁹Endogeneity (supervision influences performance which in turn influences subsequent supervision allocation decisions) is overcome by relating lagged annual supervision to annual changes in interim implementation performance whereas effectiveness is based on the final result of the project.

multiple donors will most likely yield inferior outcomes. This is true even when competition has the effect of increasing aid flows since, in this case, other objectives than poverty reduction are plausibly driving aid givers. Second, the benefits of external discipline are obtained at the price of a loss of national sovereignty in the recipient countries. In our understanding, though, the donor agency is not preoccupied by such a loss. The only thing that matters for it is the predicament of the poor whose interest entails a reduction in the power of the political elite. The donor is therefore insensitive to the latter's claim to act on behalf of the entire nation and to be in charge of guarding its sovereign prerogatives. In point of fact, such a claim is often self-serving as it helps conceal the true motives behind the elite's behavior.

Finally, the strategy consisting for rich countries of distributing aid through the channel of Non-Governmental Organizations (NGOs) is not the panacea that it is sometimes thought to be. Allocating money to NGOs or local agencies does not actually solve the problem of aid misuse altogether. In particular, NGOs or so-called local partner associations may just be shell organizations set up by unscrupulous officials who are eager to get hold of the aid money wherever it is on offer.¹⁰ Or they may be agencies unable to resist the pressures of rent-seeking officials on whom they depend for licences to operate.¹¹ Therefore, even when the donor chooses to earmark the whole aid fund for NGOs, monitoring aid use is necessary to (partially) improve governance. It is true, however, that NGOs, if only because they tend to be small and politically weak, are easier to monitor and sanction than government agencies.

Let us now turn our attention to the model itself. We consider the case of two beneficiary countries i ($= 1, 2$) where the initial, pre-aid per capita incomes of the poor are w_1 and w_2 , respectively. Their numbers are n_1 and n_2 . It is convenient to define poor people as an arbitrary fraction (say, 30 or 40 percent) of the total population making up the lower part of the income

¹⁰Thus, in the case of Benin, local NGOs and associations, which are often "empty shells established with the sole purpose of capturing aid", have multiplied within a short period of time to number several thousands (Le Monde, 26 February 2001). In non-African countries, too, NGOs often constitute "an opportunistic response of downsized bureaucrats, with no real participation or local empowerment" and, inevitably, program officers themselves become involved in the creation of community institutions (Conning and Kevane 2002: 383-84). In a more vivid language, NGOs are said to "often sprout up, like plants in the sunlight, solely to bathe in this foreign money" (Economist, Special Report: Aid to Africa, July 2-8 2005, p. 26).

¹¹Thus, a staff member of USAID, the official aid agency of the US government, recently admitted to having seen the latter scenario "repeated countless times", leading him to the conclusion that "local agencies operate at the pleasure of the host government" (Economist, May 27-June 2 2023, Letters, p. 16).

scale. Thus, n_1 and n_2 are proportional to the populations of the two countries. The donor, for instance a bilateral or multilateral development agency, has to allocate an exogenous amount of aid T between the two countries, s_1 and $s_2 (= 1 - s_1)$ being the shares of total aid going to countries 1 and 2. However, a portion γ_i of aid is devoted to the task of monitoring the way it is used, so that only a share $s_i(1 - \gamma_i)$ is actually reaching the poor in country i . This amount corresponds to the concept of 'operable' aid in the aid literature. To this subtraction we must add another that stems from local administrative cost, inefficient management or mere embezzlement of aid money by local elites. We assume that such domestic leakages represent a proportion y_i of operable aid, so that the poor of country i will eventually obtain a portion $(1 - \gamma_i)(1 - y_i)$ of the aid fund transferred to their government, and $s_i(1 - \gamma_i)(1 - y_i)$ of the total fund.

Based on these simple assumptions, the objective function of the donor is to maximize the utility derived from the improved, post-aid level of income of the poor in recipient countries, namely

$$\sum_{i=1}^2 n_i u(w_i + s_i(1 - \gamma_i)(T/n_i)(1 - y_i)) \quad (3.1)$$

where $u(\cdot)$ is a standard increasing and concave utility function. In the following, it will be convenient to use the familiar isoelastic specification:

$$u(z) = \frac{1}{1 - \epsilon} z^{1 - \epsilon} \quad (3.2)$$

where z denotes the terms between square brackets in (3.1), that is, the post-aid level of per capita income of the poor and where, as is well-known, ϵ can be interpreted as the aversion of the donor for inequality, or, in this particular case, poverty. While $1 - y_i$ is a standard measure of aid effectiveness, $(1 - \gamma_i)(1 - y_i)$ may be called a *monitoring-adjusted measure of aid effectiveness*. It reflects the role of efficiency considerations in the presence of aid monitoring.

The main contribution we seek for this paper lies in the endogenization of the leakage term y_i . It is precisely this leakage rate that the donor tries to control by spending a larger or smaller amount of resources on monitoring, possibly including sanctions in the case of excessive leakage. The leakage factor is assumed to be a function $\tilde{y}_i(\gamma_i)$, specific to each country, of γ_i , which is the expenditure on aid monitoring per dollar of aid or the unit spending

on monitoring. We reasonably assume that marginal returns to monitoring are diminishing, implying that the function $\tilde{y}_i(\gamma_i)$ is decreasing and convex. It also takes values limited to the range $[0, 1]$. We thus write:

$$\tilde{y}_i(\gamma_i) \in [0, 1]; \tilde{y}'_i(\gamma_i) \leq 0; \tilde{y}''_i(\gamma_i) \geq 0 \quad (3.3)$$

The value $\tilde{y}_i(0)$ is an indicator of the quality of what may be called the *basic (intrinsic) or the overall governance* prevailing in country i . It is self-governance consisting of either self-imposed discipline in the sense of the restraint that local elites impose on themselves, or of the internal control exerted on these elites by existing civil society organizations. We expect the leakage rate in the absence of monitoring to be high in a predatory state and low in a developmental state. Whichever the situation, the point is that through monitoring donors are able to improve the governance of aid programs with a view to making them more effective than other public funds targeting the poor, which are subject to only basic governance. A direct implication is that the worse a country is governed, the larger the share of the aid budget the donor should be tempted to devote to its monitoring. By the same token, badly governed or fragile countries would receive lower amounts of (per capita) operable aid, so that poor people will be worse off on that count alone. How the donor resolves such a trade-off for each country will affect the way aid is allocated between the better and the worse-governed countries.

To make the meaning of the function $\tilde{y}_i(\gamma_i)$ more explicit and to facilitate the resolution of the model, it is convenient to use a particular specification. The rest of this paper will be based on the following simple functional form:

$$\tilde{y}_i(\gamma_i) = \lambda(\beta_i, \gamma_i) = \frac{1}{\beta_i + \varphi(\gamma_i)} \text{ with } \beta_i > 1; \varphi(\gamma_i) = b\gamma_i^{1-\alpha} \quad \alpha \in [0, 1] \quad (3.4)$$

Clearly this function satisfies the properties (3.3). $\beta_i (= 1/\tilde{y}_i(0))$ may be interpreted as an indicator of the *basic or overall governance quality* in country i , i.e. the level of governance observed in local public projects in general or, more precisely, the capacity to actually channel public funds directly or indirectly towards poverty reduction. The additive expression $\beta_i + \varphi(\gamma_i)$ reflects the fact that the donor may improve the country's basic governance when it comes to managing an aid program. Therefore, $\varphi(\gamma_i)$ may be interpreted as a kind of external

control imposed by the donor in order to make aid leakages more difficult or costly for the government, and thus improve the poverty-reduction effectiveness of aid funds. This is done through close monitoring of aid projects and programs and the imposition of sanctions in case of proved embezzlement. It is plausibly an increasing and concave function of the amount spent on monitoring per unit of aid, γ_i . Added to the exogenously given, country-specific basic governance β_i , the level of external discipline determines the aggregate level of governance quality applied to aid use in country i . Note that the function $\varphi(\gamma_i)$ may implicitly account for a possible reaction of the local authorities to the discipline imposed by the donor, plausibly in the form of a reduction of their own monitoring effort applied to aid projects. This will be discussed further in Section 5.

The model of aid allocation under endogenous, aid-specific governance is now complete. The donor must choose four key parameters: the allocation of total aid among the two countries, s_1, s_2 , and the rate of monitoring in the two countries, γ_1 and γ_2 . They are set in such a way as to maximize the objective function (3.1), under the constraints (3.4) and $s_1 + s_2 = 1$.

The expression (3.4) may look rather ad hoc, although it conveys an important intuition correctly. In fact, a solid micro-foundation exists for it, which separates the respective roles of the probability of detection of aid-related fraud and the punishment meted out in the event of fraud detection. Moreover, (3.4) can be analytically derived as the outcome of a Nash bargaining game between a local elite and the intended beneficiaries (the poor). In this bargaining setup, basic discipline β_i stands for the relative bargaining power -via the reservation utilities- of the latter vis-à-vis the former. In short, (3.4) can be conceived as being drawn from a structural model sufficiently general to allow for multiple interpretations of the underlying mechanism of aid embezzlement, which can depend upon the context we have in mind. In particular, it can accommodate a situation in which aid flows pass through an intermediary acting as a screen between the donor and the beneficiaries. In order to focus on the main task of the paper, namely aid allocation between two countries when endogenous discipline is allowed, we do not enter here into more details about the microeconomics behind (3.4). For that purpose, we refer the interested reader to a previous work (see Bourguignon and Platteau, 2018) in which these details are provided in the framework of a single recipient country.

It bears noticing that in the above formulation the cost of monitoring aid use is imputed to each country in proportion of the amount of monitoring effort dedicated by the donor. An

alternative approach would consist of viewing monitoring expenses as a cost incurred by the donor in addition to the aid funds themselves. The donor's objective function would then be written as:

$$\sum_{i=1}^2 n_i u(w_i + s_i(T/n_i)(1 - y_i)) - \sum_{i=1}^2 s_i T \gamma_i$$

We did solve this version of the model, but the exercise turned out to be much more intricate than the analytical work based on the country-specific imputation of monitoring costs behind (3.1). This is due to the lack of separability between aid shares and monitoring intensity in the optimization of the objective function, unlike what is obtained with (3.1). Results are essentially of the same nature under both approaches, but much simpler with separability, hence our decision to present only the simpler version in this paper. The interested reader can obtain the more cumbersome, alternative version upon request from the authors.

In what follows, we first characterize the solution of the model and then proceed by analyzing some key properties, most importantly the way aid allocation between the two countries and the donor's monitoring effort for each country depend on the basic governance levels, β_1 and β_2 , and on the pre-aid incomes of the poor in the two countries. Without loss of generality, it is assumed in the rest of this paper that $\beta_1 < \beta_2$. In other words, country 1 is supposed to be the worse-governed country. In agreement with the observation made in the introduction that countries with weak governance are often the poorest, we make the additional assumption that the poor people of country 1 have a lower level of living than in country 2: $w_1 < w_2$. We shall stress the instances where the latter assumption matters for the validity of a proposition.

3.2 Solution of the model

The separability of the objective function (3.1) with respect to γ_i considerably simplifies its maximization. Differentiating first with respect to γ_i , and using the specification $\lambda(.,.)$ of the leakage rate in (3.4), we obtain the following first-order condition for maximization of (3.2).¹²

$$-(1 - \gamma_i) \lambda_{\gamma}^i = 1 - \lambda^i \quad (3.5)$$

where λ^i is a shortcut for $\lambda(\beta_i, \gamma_i)$ and λ_{γ}^i stands for the partial derivative of $\lambda(.,.)$ with

¹²Note that the first-order condition writes $n_i (z_i^e)^{-1} z_{\gamma_i} = 0$, implying $z_{\gamma_i} = 0$.

respect to argument $X = (\beta_i, \gamma_i)$ for $i = 1, 2$. Multiplying both sides by $s_i T$, this equation states that at equilibrium the marginal benefit of reducing the leakage so that more operable aid reaches the poor should be equal to the marginal cost corresponding to the required increase in monitoring. Equation (3.5) thus gives the optimal value of unit spending on monitoring as a function of the country's basic governance, β_i , and the effectiveness of that spending in reducing the leakage as summarized by the b and α parameters. Due to the separability of the objective function (3.1) with respect to γ_i , the optimal value of monitoring is independent of the pre-aid income of the poor, which is a very convenient property. Furthermore, it is easily seen that the second-order condition for maximization is satisfied because of the convexity of the function $\lambda(\beta_i, \gamma_i)$ with respect to γ_i .

Taking the derivative of the objective function with respect to s_1 , and making use of $s_2 = 1 - s_1$, yields the following condition:

$$\frac{u'_1}{u'_2} = \frac{(1 - \tilde{y}_2(\gamma_2))(1 - \gamma_2)}{(1 - \tilde{y}_1(\gamma_1))(1 - \gamma_1)} \quad (3.6)$$

where u'_i stands for the marginal utility the donor derives from the net post-aid income of the poor in country i . This condition is the familiar equality between the rate of substitution of the donor's preferences between the poor in the two recipient countries and the relative effectiveness in transferring income to them. This relative effectiveness is known once the first-order conditions (3.5) on γ_i have been solved. Equation (3.6) is then an equation in s_1 , which appears in the arguments of the marginal utilities u'_1 and u'_2 . In the case of the iso-elastic specification (3.2) of the utility function, it is possible to get an analytical expression of its solution:

$$\left(\frac{w_1 + s_1(1 - \gamma_1^*)(T/n_1)(1 - \lambda(\beta_1, \gamma_1^*))}{w_2 + s_2(1 - \gamma_2^*)(T/n_2)(1 - \lambda(\beta_2, \gamma_2^*))} \right)^{-\varepsilon} = \frac{(1 - \lambda(\beta_2, \gamma_2^*))(1 - \gamma_2^*)}{(1 - \lambda(\beta_1, \gamma_1^*))(1 - \gamma_1^*)} = \frac{E_2^*}{E_1^*} \quad (3.7)$$

where γ_i^* is the optimal value of unit spending on monitoring obtained from (3.5), and E_i^* is the optimized monitoring-adjusted aid effectiveness in country i . As has been mentioned earlier, the latter measure corresponds to standard aid effectiveness, $1 - \tilde{y}_i(\gamma_i^*)$, corrected by the operable aid ratio, $1 - \gamma_i^*$. Solving for s_1 and making use of (3.2), it comes easily that an interior solution in the $]0, 1[$ interval is given by:

$$s_1 = \frac{(E_1^*/E_2^*)^{1/\varepsilon}(w_2 + E_2^*T/n_2) - w_1}{T(E_1^*/n_1 + (E_1^*/E_2^*)^{1/\varepsilon}E_2^*/n_2)} \quad (3.8)$$

This expression may be rewritten so as to make apparent the role of the 'need-adjusted aid effectiveness' concept $E_i^{*1/\varepsilon}/w_i$, as used in Bourguignon and Platteau (2021):

$$s_1 = \frac{E_2^{*1-1/\varepsilon}/n_2}{E_1^{*1-1/\varepsilon}/n_1 + E_2^{*1-1/\varepsilon}/n_2} + \frac{w_2 E_2^{*-1/\varepsilon} - w_1 E_1^{*-1/\varepsilon}}{T (E_1^{*1-1/\varepsilon}/n_1 + E_2^{*1-1/\varepsilon}/n_2)} \quad (3.9)$$

Thus, the aid share of a country increases with its need-adjusted aid effectiveness, while it decreases with that of the other country. On the other hand, the first term on the right-hand side shows how the aid share depends on demographic weights corrected by the corresponding levels of monitoring-adjusted aid effectiveness.

3.3 Properties of the solution

It bears emphasis that, once the optimal monitoring efforts are applied, monitoring-adjusted aid effectiveness (E_i) is fixed in both countries, and the properties of the aid allocation model are identical to those derived in the case of exogenous aid effectiveness (see Bourguignon and Platteau, 2021). They are briefly listed in what follows.

To begin with, it is straightforward that the share of country 1 decreases with w_1 and increases with w_2 . Much less obvious is the effect of the volume of aid, T , on aid shares, since it turns out to depend on the relative values of *need-adjusted aid effectiveness*, $E_i^{1/\varepsilon}/w_i$ in the two countries, as is evident from (3.9). Paradoxically, the share of the most need-adjusted aid effective country falls with the volume of aid, a property which requires the worse-governed country to be also poorer prior to the aid transfer. What happens is that aid makes post-aid incomes less unequal, implying that an increase in total aid available has the effect of raising the post-aid income of the poor to a greater extent in the worse-governed and poorer country than in the other, better-governed and richer country. As a result, the marginal attractiveness of the former country to the donor is comparatively reduced so that restoration of equilibrium necessitates that its aid share be lowered.

Finally, and even more intriguingly, it has been proven in Bourguignon and Platteau (2021) that the share of aid going to country 1 would increase with its aid effectiveness, E_1 , yet only if the poverty aversion of the donor is below some threshold value above unity. It would fall if poverty aversion is above that threshold and, in particular, if it becomes ex-

tremely large and close to the Rawlsian criterion. Because they will be used again in the sequel of this paper, both this property and its proof are worth stating in full.

Proposition 1. *There exists a critical value of poverty aversion such that the aid share of the worse-governed country increases with its basic governance below this value and decreases above it.*

Proof. It relies on the following property of the optimal aid share:

$$\text{sign} \left\{ \frac{\partial s_1}{\partial E_1^*} \right\} = \text{sign} \{ 1 - \mu_1 \varepsilon \} \quad (3.10)$$

where μ_1 is the share of aid in the post-aid income of the poor in country 1, that is, $s_1 TE_1^* / (w_1 + s_1 TE_1^*)$. This property follows directly from the first-order condition (3.7), which may be rewritten as:

□

$$\left(\frac{w_1 + s_1 TE_1^*}{w_2 + s_2 TE_2^*} \right)^{-\varepsilon} = \frac{E_2^*}{E_1^*}$$

The logarithmic derivative of both sides with respect to E_1 are respectively $-\mu_1 \varepsilon$ and -1 , hence the sign relationship (3.10). In addition, it is easily seen that μ_1 is increasing with s_1 which, according to (3.8), is itself an increasing function of ε because $E_1^*/E_2^* < 1$. The expression on the right-hand side of (3.10) is, therefore, a monotonically decreasing function of ε , and it goes from 1 when ε tends towards 0 to negative values when ε goes to infinity. As for s_1 , it remains strictly positive. QED The intuition behind the latter findings deserves to be highlighted because it helps us to grasp the complexity of the problem at stake. The key fact is that two opposite effects are at work, and whether one dominates the other depends on the value of ε . Through the first, a substitution effect, the donor is induced to reward better governance by dedicating a higher aid share to the country whose basic governance has improved. The second effect, an income effect, stems from the fact that the income of the poor in that country has increased relative to the income of the poor in the other country. On that count, the donor wants to reduce the aid share awarded to the country whose governance

has improved. The more poverty-averse the donor, the more sensitive to the income effect. Therefore, when poverty aversion is strong enough, the second effect outweighs the first and an inverse relationship between aid effectiveness and aid share is obtained. Because of the absence of any income effect in the PBA aid allocation rules, a country aid share can only move in tandem with its governance quality (or aid effectiveness) when these rules are implemented.

Having uncovered the complex working of the aid allocation model and the nature of the role of poverty aversion in the simple context of exogenous aid effectiveness, we are now ready to turn to the core issue of the present paper, namely how the ability of the donor to influence the governance of aid use plays out in the same setup of one donor and two heterogeneous recipient countries. In particular, we want to know how monitoring-adjusted aid effectiveness, E_i , varies with the basic governance of country i , β_i . The answer to this question is stated in Proposition 2.

Proposition 2. *The optimal unit spending on monitoring by the donor falls with the basic governance of a country, implying that basic governance and external monitoring are substitutes. Although in the opposite direction, the donor's reaction is never strong enough to cause an increase in the aid leakage rate. As a result, the monitoring-adjusted aid effectiveness of a recipient country is an increasing function of its basic governance.*

Proof. Differentiating (3.5) with respect to β_i and γ_i yields:

$$\frac{\partial \gamma_i}{\partial \beta_i} = -\frac{\lambda_{\beta}^i - (1 - \gamma_i)\lambda_{\beta\gamma}^i}{2\lambda_{\gamma}^i - (1 - \gamma_i)\lambda_{\gamma\gamma}^i}$$

where both the denominator and the numerator are clearly negative, so that $\partial \gamma_i / \partial \beta_i$ is unambiguously negative. To show that the leakage rate also falls with β_i , note that $\lambda_{\gamma}^i = -\lambda^{i^2} \phi'(\gamma_i)$ so that (3.5) may be rewritten as:

$$(1 - \gamma_i)\phi'(\gamma_i)\lambda^{i^2} = 1 - \lambda^i \quad (3.11)$$

Consider now an increase in β_i and a drop in γ_i that leave λ^i unchanged. As the left-hand side of the preceding equation is a decreasing function of γ_i , it has now become greater than the right-hand side. Re-establishing equilibrium thus requires lowering λ^i . It follows that, at the optimum level of monitoring, the extent of aid leakage declines as the country's basic governance (or aid effectiveness) improves, although somewhat less than if the drop in unit spending on monitoring had not occurred. In any case, since both standard aid effectiveness, $1 - \lambda^i$, and the operable aid share, $1 - \gamma_i$, increase, so does monitoring-adjusted aid effectiveness, E_i . QED □

Proposition 2 thus says that the donor will apply more monitoring to the worse-governed country, country 1, yet not to such an extent that it could make the magnitude of aid leakage smaller, and monitoring-adjusted aid effectiveness (E_1) higher, than in country 2. The next and most important step of our analysis consists of comparing the regimes of endogenous and exogenous discipline by looking at the end outcomes in terms of the welfare of the poor in the two recipient countries and the pattern of aid allocation between them.

4 Comparing aid allocation under endogenous and exogenous or uniform monitoring

4.1 The issues at stake

This section is based on the observation that the possibility of varying the extent of monitoring when deciding about the allocation of aid among different countries does not seem present in the explicit or implicit formulae used today by bilateral or multi-lateral development agencies. The idea is to evaluate how different their aid allocation would be if they were optimizing aid monitoring based on their knowledge of the basic governance of each recipient country.

An extreme case would be to assume that development agencies play no role in the way

aid funds are being used. They sign a check and let the recipient countries' government manage the corresponding funds as it wishes. In practice, however, some measure of monitoring takes place, which is codified in the bureaucratic functioning of these agencies. In many instances, it is even the case that an evaluation is undertaken at the end of projects or programs to show to their constituencies how aid money was used. This evaluation is admittedly based on some form of information collected and monitoring applied throughout the life of the funded projects and programs. None the less, it is typically run in a modest or even perfunctory manner, meaning that it relies on rather standardized reports written by project or program managers rather than on detailed inquiries led by an independent staff dedicated to the task. Some agencies may be more ambitious than others and exert a tighter monitoring, yet their operations are always part of a bureaucratic routine. There is thus ground to believe that the intensity of monitoring is rather uniform among recipient countries, instead of being country-specific as in the model presented in Section 3.

Based on such a scenario, the present section compares the impact of aid and its allocation across countries under two different monitoring scenarios: (i) an exogenous, uniform monitoring intensity, γ^0 , applied to all the recipient countries; (ii) an optimal, or endogenous, country-specific monitoring intensity, γ_i^* . Under (i), two particular cases are of interest: the case of no monitoring where $\gamma^0 = 0$, and the case where the better-governed country is considered as the reference point by the donor, who thus sets the exogenous uniform monitoring intensity at the level that would be optimal for that country. Since it is assumed that $\beta_1 < \beta_2$, this means that $\gamma^0 = \gamma_2^*$. Practically, however, it turns out that results are qualitatively similar for all values of the exogenous uniform monitoring intensity in the interval $[0, \gamma_2^*]$. This is the reason why we do not distinguish between the two particular cases in the following analysis.

Three sets of questions need to be elucidated. First, what is the impact of optimizing the intensity of monitoring on relative levels of aid effectiveness, and how poor people in the two countries benefit from this change? Second, how do aid shares change when the donor shifts from exogenous to endogenous discipline (or monitoring)? Finally, is the sensitivity of the aid allocation rule to pre-aid incomes and basic governance parameters modified when the same shift of monitoring regimes occurs? As pointed out above, we consider that exogenous uniform monitoring is implicitly the current *modus operandi* of many development agencies while country-specific monitoring is the optimal procedure that all donors ought to follow.

Seen in this light, the questions that we raise are of great policy relevance.

To simplify notations and derivations, we henceforth assume that $n_1 = n_2 = 1$. This is an innocuous assumption insofar as all the following propositions would remain valid in the general case.

4.2 The efficiencywise superiority of endogenous monitoring

The first issue when we compare endogenous, country-specific monitoring with uniform monitoring is how the two regimes can be ranked in terms of monitoring-adjusted aid effectiveness in the two countries. The result of the comparison is stated in Proposition 3.

Proposition 3. *Compared to the case where the intensity of monitoring is exogenous and uniform in the interval $[0, \gamma_2^*]$, endogenous monitoring causes monitoring-adjusted aid effectiveness to be higher in the two countries but relatively more so in the worse-governed country.*

Proof. The gain in effectiveness when moving from exogenous to optimal monitoring in country i is given by:

$$G_i = \frac{(1 - \gamma_i^*)(1 - \lambda(\beta_i, \gamma_i^*))}{(1 - \gamma^0)(1 - \lambda(\beta_i, \gamma^0))} = \frac{E_i^*}{E_i^0}$$

where E_i^0 denotes aid effectiveness in country i when monitoring is exogenously determined. Differentiating this expression with respect to β_i and making use of the first-order maximization condition (3.5), the sign of that derivative is given by:

$$\text{sign} \left(\frac{dG_i}{d\beta_i} \right) = \text{sign} \left\{ -\frac{\lambda_\beta(\beta_i, \gamma_i^*)}{1 - \lambda(\beta_i, \gamma_i^*)} + \frac{\lambda_\beta(\beta_i, \gamma^0)}{1 - \lambda(\beta_i, \gamma^0)} \right\}$$

Bearing in mind that $\lambda_\beta = -\lambda^2$ and that the function $f(x) = x^2/(1 - x)$ is increasing in x if $x < 1$, it turns out that $dG_i/d\beta_i$ is unambiguously negative since $\lambda(\beta_i, \gamma_i^*) \leq \lambda(\beta_i, \gamma^0) < 1$. The relative gain in (monitoring-adjusted) aid effectiveness due to endogenous monitoring is thus higher for the worse-governed country. Indeed, since $\beta_2 > \beta_1$, we have that $E_1^*/E_1^0 > E_2^*/E_2^0$

and, therefore,

$$\frac{E_1^*}{E_2^*} > \frac{E_1^0}{E_2^0} \quad (4.1)$$

which implies $\frac{E_1^*}{E_1^0} > \frac{E_2^*}{E_2^0}$. QED □

The general increase in aid effectiveness caused by the use of optimal tailor-made monitoring is an expected result, and it is linked to the separability of the donor's objective function with respect to unit monitoring spending. The intuition behind the larger increase observed for the worse-governed country is essentially that the restriction of the exogenous uniform monitoring, γ^0 to the interval $[0, \gamma_2^*]$, together with Proposition 2, imply that $\gamma^0 \leq \gamma_2^* < \gamma_1^*$. The endogenization of monitoring thus leads to a bigger change in the worse-governed country.

The comparatively strong increase in monitoring-adjusted aid effectiveness entails a similar property for the post-aid income of the poor, aid shares being constant. This additional result necessarily holds when the pre-aid income is lower in the worse-governed and poorer country. The reason is that aid represents a higher share of the poor's post-aid income in that country, so that it has the effect of bridging part of the inter-country gap in the poor's incomes.

4.3 Comparing aid shares under endogenous and exogenous monitoring

Our task under this subsection is to examine the effect on aid shares caused by a shift from exogenous uniform monitoring to optimal country-specific monitoring. In a first stage, we consider the case of corner solutions, ignored until now, and the possibility that the endogenization of monitoring makes a country aid-worthy in the eyes of a donor who did not deem it eligible under exogenous uniform monitoring. Clinging to our assumption that country 1 is the worse-governed recipient ($\beta_1 < \beta_2$), it can be seen from (3.8) that, in terms of monitoring-adjusted aid effectiveness, the condition under which this country will receive no aid ($s_1 = 0$)

is:

$$E_1^{*1/\varepsilon}/w_1 \leq \frac{E_2^{*1/\varepsilon}/w_2}{1 + TE_2/w_2}$$

In words, country 1 receives no aid if its need-adjusted aid effectiveness is smaller than a function of the same indicator in the other country. From the proof of Proposition 3, we know that $E_1^*/E_1^0 > E_2^*/E_2^0$. It follows that a switch from exogenous to endogenous monitoring would raise the value of the left-hand side of the preceding condition more than the value of the right-hand side, thus making the condition harder to satisfy. Therefore, if we assume that the above condition holds under exogenous monitoring, it is quite possible to encounter situations in which it is violated under endogenous monitoring. Hence the following proposition:

Proposition 4. *The donor's shift from exogenous uniform to endogenous monitoring may cause the worse-governed country to become aid-eligible.*

We can now tackle the case of an interior solution. The way the aid share of the worse-governed country compares under the two regimes is stated in Proposition 5, the proof of which is provided in Appendix A. The critical role of the degree of the donor's poverty aversion clearly comes out.

Proposition 5. *If the poverty aversion of the donor lies above some floor value smaller than unity, and if the exogenous, uniform unit spending in monitoring, γ^0 , lies in the interval $[0, \gamma_2^*]$, there then exists a threshold on the pre-aid income of the poor in the worse-governed country below which that country's aid share becomes smaller when the donor switches from exogenous to endogenous monitoring. Above the threshold, which is an increasing function of the pre-aid income in the better-governed country, the aid share of the worse-governed country becomes larger.*

The logic behind Proposition 5 is revealing of the way our model works. We know that the switch from exogenous (uniform) to endogenous (country-specific) monitoring makes

aid relatively more effective -in the sense of monitoring-adjusted effectiveness- in the worse-governed country, which prompts the donor to raise its aid share, according to a standard substitution mechanism. However, the same switch also increases the ratio of the post-aid incomes of the poor in favor of the worse-governed country. This sparks an income effect, which is more or less pronounced depending on the relative amounts of initial pre-aid incomes. Clearly, the switch of monitoring regime will entail an increase in the aid share of the worse-governed country if the substitution effect outweighs the income effect, keeping in mind that the strength of the latter effect depends on the degree of the donor's poverty aversion. Yet, for a given aversion, the size of the income effect also depends on the pre-aid income of the poor in the worse-governed country. If that income is large relative to the other country, the income effect is dominated by the substitution effect and the monitoring switch raises the aid share of the worse-governed country. The opposite outcome is obtained if the pre-aid income of the poor in the same country is small, giving rise to a large income effect.

Finally, the Rawlsian case where poverty aversion tends towards infinity provides an illustration of the limit case in which the aid share allocated to the worse-governed country falls as a result of the regime shift. In this case, the post-aid incomes of the poor are identical in the two countries:

$$w_1 + s_1 E_1^* T = w_2 + (1 - s_1) E_2^* T$$

Solving for s_1 and comparing the exogenous and endogenous monitoring regimes yields:

$$\Delta s_1 = \frac{w_2 - w_1}{T} \cdot \left(\frac{1}{E_1^* + E_2^*} - \frac{1}{E_1^0 + E_2^0} \right) + \left(\frac{E_2^*}{E_1^* + E_2^*} - \frac{E_2^0}{E_1^0 + E_2^0} \right)$$

Since the two terms in square brackets are negative, the whole expression is also negative for w_1 smaller than some threshold above w_2 , which is in agreement with Proposition 5.

4.4 Post-aid incomes of the poor under endogenous and exogenous monitoring

More important than the aid shares accruing to individual recipient countries, which depend on their basic governance and the incomes of their poor, is the net change in these incomes after aid has been delivered. Putting Propositions 3 and 5 together, it appears that the post-aid income of the poor in the worse-governed country unambiguously increases when the donor shifts to tailor-made optimal monitoring. However, this result is not guaranteed in the case of the better-governed country. The source of ambiguity lies in the conjunction of two effects. On the one hand, it was seen earlier that, the aid shares being given, the shift has the effect of raising the levels of monitoring-adjusted aid effectiveness in the two countries, and hence of the post-aid incomes of their poor, but relatively more so in the worse-governed and poorer country. On the other hand, aid shares are modified. When the share allocated to the worse-governed country increases, it reinforces the aid effectiveness effect in that country while it runs against it in the better-governed country, opening the possibility that the post-aid income of the poor declines in the latter. In principle, the situation could be reversed when the aid share of the worse-governed country falls with the shift of monitoring regime. We can nevertheless show that this outcome will never arise, essentially because the aid effectiveness effect is relatively more pronounced in that country. The following proposition, the proof of which is presented in Appendix B, summarizes these properties.

Proposition 6. *When switching from exogenous, uniform monitoring γ^0 in the interval $[0, \gamma_2^*]$, to country-specific endogenous monitoring, the post-aid income of the poor in the worse-governed country increases relative to the corresponding income in the other country. This also holds true in absolute terms. As for the the post-aid income of the poor in the better-governed country, it may fall.*

It is evident that the conditions under which the post-aid income of the poor in the better-governed country falls when the donor switches to endogenous monitoring depend on the unit spending on monitoring, γ^0 , which is exogenously given. In this respect, a trivial but informative property of the model is that the post-aid income of the poor in the better-governed

country necessarily decreases with monitoring endogenization if γ^0 is close to the optimal monitoring estimated for that country, γ_2^* . In such a case, indeed, the monitoring-adjusted aid effectiveness for country 2 does not change perceptibly with the switch to tailor-made monitoring, yet its aid share falls under the conditions of Property 5. The post-aid income of the poor in the better-governed country therefore ends up being lower than under exogenous monitoring.

4.5 Sensitivity of aid shares under endogenous and exogenous monitoring

There is a last problem that we want to address. It concerns the comparative sensitivity of the aid allocation pattern to pre-aid incomes and basic governance under the regimes of endogenous and exogenous monitoring. This is a topical issue because we would like to know how the aid allocation (PBA) rules used by many donor agencies are biased. More precisely, do these rules put an excessive weight on the basic governance or on the income level of the poor in the recipient countries? As seen below, there is no clear answer to that question, except in one single instance.

Our approach is based on a comparison between the relevant partial derivatives of the aid share of the worse-governed country under endogenous and exogenous monitoring. When sensitivity to variations in pre-aid income of the poor is considered, the exercise yields the results stated in Proposition 7 and proven in Appendix C.

Proposition 7. *In comparison with the situation obtained under exogenous uniform monitoring, aid allocated to the worse-governed country under optimal monitoring is less sensitive to variations in the pre-aid per capita income of its poor. This is also true of the sensitivity to variations in the pre-aid income of the poor in the other country, yet only if the poverty aversion of the donor is below some threshold smaller than unity.*

In other words, endogenous monitoring acts as a kind of shock-absorber that cushions the

impact of pre-aid income variations on the aid share of the worse-governed country: that aid share will be less affected by a change in pre-aid income than it would be under a regime of exogenous (uniform) monitoring. This is not generally true when the change in the poor's pre-aid income occurs in the better-governed country.

The analytics are more intricate when comparing the sensitivity of the aid share of the worse-governed country to its basic governance under the two monitoring regimes. This complexity reflects the non-intuitive result, recalled in Proposition 1, according to which this aid share may well be decreasing with the same country's level of aid effectiveness. The consequence of the non-monotonous relationship between basic governance and aid share for the worse-governed and poorer country is that no property similar to the one obtained in Proposition 7 can be established regarding variations in basic governance. One conclusion can safely be drawn, though: the direction of change of the aid share when the basic governance of the worse-governed country varies may not be the same under the two monitoring regimes. To see this, let us start by stating the following lemma (the proof is provided in Appendix D):

Lemma 1. *Let ε^0 and ε^* be the thresholds of the poverty aversion parameter above which the optimal aid share of the worse-governed country starts decreasing with its basic governance under exogenous (uniform) and endogenous monitoring, respectively. Then, it is the case that $\varepsilon^* < \varepsilon^0$.*

Then, an interesting proposition follows:

Proposition 8. *There exists a range of values of the poverty aversion parameter such that the aid share of the worse-governed country increases with its basic governance under exogenous uniform monitoring, but decreases under endogenous monitoring.*

The intuition behind Proposition 8 is easily elucidated. We just need to bear in mind the mechanism behind Proposition 5: the income effect sparked by an improvement in the basic governance of the worse-governed country is necessarily larger under endogenous monitoring, precisely because monitoring is optimized and aid represents a larger share of the poor's post-aid income. Since, as we know, the income effect prompts the donor to lower that country's aid share (especially so if poverty aversion is strong), we expect a comparatively sharp

decrease under endogenous monitoring. Note carefully that this outcome is obtained because exogenous monitoring is assumed to be at a level at most equal to the optimal monitoring intensity for the better-governed country, which is itself smaller than the optimal intensity for the worse-governed country.

5 Discussion and extensions

This section discusses several possible extensions of the model proposed in the preceding sections, and makes a number of additional comments.

The first issue is about the possible reaction of the recipient country to the discipline imposed by the donor to improve the effectiveness of aid use. It has been mentioned earlier that the specification of the model is actually consistent with such a reaction insofar as it consists of a relaxation, by the recipient country's elite, of its own (basic or overall) discipline when running projects and programs funded by aid transfers. It is only a matter of interpretation of the coefficient α : the value of this parameter may reflect both the monitoring technology of the donor and, in the opposite direction, the governance-related reaction of the aid-receiving central or local governments. Presumably, this is not very different from assuming that the basic governance parameter, β_i , may fall in response to external discipline. Going beyond this simple way of endogenizing the recipient country's reaction, and assuming that the disciplining intervention of donor truly modifies the basic governance in that country, would logically involve making the pre-aid income of the poor, w_i , dependent on the quality of the basic governance parameter, β_i . In other words, the discipline imposed by the donor would trigger a reduction of the poor's pre-aid income in the recipient country. This seems to be a rather extreme assumption, though.¹³

A second issue concerns the tailoring of monitoring effort to the specific situation of each recipient country. There are serious reasons why a donor may prefer uniform to differentiated monitoring. In particular, equality of treatment of heterogeneous countries may be

¹³Note that making w_i an increasing function of β_i in our derivation work would essentially reinforce the income effect of an improvement in basic governance. This would most likely lower the poverty aversion threshold above which the aid share starts decreasing with basic governance.

more palatable or justifiable, especially so if differences in governance quality are largely attributable to historical or other factors beyond the control of present-day governments. In this case, a hard trade-off arises between equity and efficiency considerations. The case of exogenous uniform monitoring at a level below what would be optimal in either recipient country has been analyzed in the previous section. Considering now the case of uniform monitoring when its intensity is optimized by the donor, the following maximization problem must be solved::

$$\text{Max}_{s_1, s_2, \gamma^0} u(w_1 + s_1 T(1 - \gamma^0)(1 - y_1)) + u(w_2 + s_2 T(1 - \gamma^0)(1 - y_2)) \text{ s.t. } s_1 + s_2 = 1$$

where the simplifying assumption $n_1 = n_2 = 1$ is retained. It is shown in the Appendix to this paper that the solution, γ^{0*} , of this maximization problem is such that:

$$\gamma_2^* \leq \gamma^{0*} \leq \gamma_1^*$$

where γ_i^* is the optimal monitoring intensity applied to country i when differentiated monitoring is allowed. As could be expected, Propositions 3 to 7 above remain valid if γ^{0*} is sufficiently close to γ_2^* , while they would not hold anymore if γ^{0*} gets close to γ_1^* .

In short, optimizing the uniform intensity of monitoring instead of considering it as fully exogenous, and anywhere below the optimal intensity for the better-governed country, makes the effect on aid shares of optimal country-tailored monitoring even more ambiguous. However, since the direction of variation shifts when the optimal uniform monitoring intensity goes from its optimal value for the better-governed country to the optimal value for the worse-governed county, a configuration of the parameters of the model must exist in the neighborhood of which the aid share is rather insensitive to the type of optimal monitoring regime.

A third variation around the basic model consists of extending it to any number of countries. This is a rather trivial exercise when using the model based on country-specific monitoring. The reason is that the optimal level of monitoring does not then depend on the income of the country's poor, so that it does not depend either on its aid share and the aid volume received. Properties 3 to 7 generalize by comparing each recipient country to some kind of mean over all other countries with a strictly positive aid share. Things are radically different in the case of optimal uniform monitoring intensity: it is now the average of the marginal gain

from monitoring obtained for all the recipient countries and duly weighted by their respective aid shares, that must be set to zero, instead of the marginal gain for each country taken separately (as it is the case under differentiated monitoring). The analytical characterization of the optimal common intensity of monitoring is quite intricate, even more so as the set of countries with strictly positive aid shares is itself to be determined. However, conclusions analogous to those reached in the two-country model when optimal uniform monitoring is compared with optimal country-specific monitoring are likely to hold once the situation in each recipient country is related to an average computed for all the other countries.

6 Numerical simulations

Before concluding, it is useful to illustrate our main results with the help of numerical simulations based on set values for the key parameters of country 2 (w_2 and β_2). The calibration of the model is performed under the following assumptions. Considering country 2 as the reference country, we examine how aid allocation changes with the characteristics of country 1, assuming that its overall governance is worse than in country 2: $\beta_1 \leq \beta_2$. More precisely, we define the poor as the 40 percent poorest individuals in a country, and we arbitrarily fix $w_2 = 100$ and $\beta_2 = 5$. The latter assumption about β_2 implies that the extent of aid leakage (the proportion of aid that does not reach the poor) in country 2, y_2 , is 20 percent in the absence of monitoring. Given that the aid to GDP ratio for Sub-Saharan African IDA countries is 3.6 percent,¹⁴ it can be estimated that the income of the poorest 40 percent people would be raised by 30 percent if they were to receive, directly or indirectly, the whole aid transfer accruing to their country (that is, if the aid leakage were zero).¹⁵ If the leakage is 20 percent, the poor's income would thus be raised by only $0.30(1-0.2) \times 100 = 24$ percent. This increase may still appear to be big, albeit not implausible when considering the portion of aid spending

¹⁴IDA (International Development Agency) is the aid arm of the World Bank. Only low-income and some lower-middle income countries are eligible to IDA aid.

¹⁵The 30 per cent figure is obtained by assuming that household incomes represent roughly two-thirds of GDP while, on average, the poorest 40 per cent of the population get 18 per cent of the total household incomes in Sub-Saharan Africa.

that reaches the poor indirectly, that is, through general purpose spending or investment.¹⁶

Moreover, we assume that $n_1 = n_2 = 1$, meaning that the size of the population, or the number of the poor (since poor people represent 40 percent of the population), is the same in the two aid recipient countries. What differs is the severity of poverty, as measured by w_i . In the following simulation exercise, w_1 will be allowed to vary from 50 to 110, equivalent to a variation from .5 to 1.1 times the level of w_2 . Finally, the baseline values of the cost function parameters common to both recipient countries are: $\alpha = .5$ and a value for b such that the optimization procedure delivers interior solutions in most of the range of variation of the parameters w_1 and β_1 .

Four sets of graphs are shown here. In all of them, the pre-aid income of the poor in country 1 (the worse-governed) is measured along the horizontal axis. Measured along the vertical axis are that country's aid share in Fig. 1 and 2 and the post-aid income of poor people in the two countries in Fig. 3 and 4. What distinguishes Fig. 1 from Fig. 2 and Fig. 3 from Fig. 4 is the assumed value of the donor's poverty aversion: it is equal to 1 in Fig. 1 and 3, and to 3 in Fig. 2 and 4. In Fig. 1 and 2, on the other hand, four pairs of curves are depicted, which correspond to four different values of basic governance in country 1: while β_2 is fixed and equal to 5, β_1 takes on the successive values of 1.5, 2, 3, and 5. Under the last assumption, the two countries are thus assumed to have the same level of basic governance. For a given value of β_1 , a pair of curves allows a comparison between the situations obtained under endogenous monitoring (see the solid curves) and those obtained under exogenous uniform monitoring (the dashed curves), assuming zero monitoring in the latter case.

In Fig. 3 and 4, by contrast, only one value has been selected for the basic governance prevailing in country 1 ($\beta_1 = 2$). The use of solid and dashed curves to represent the exogenous and endogenous monitoring regimes is maintained but, in these figures, the impact on the post-aid income of the poor under either regime is estimated not only for country 1 but also for country 2.

¹⁶To check that a 24 percent increase in income is not disproportionate, we may notice that transfers in cash and kind to the poor (on account of education and health) represent, on average, a bit less than 20 percent of the income of the poorest 40 percent in the three Sub-Saharan countries that appear in the Commitment to Equity Institute database (<https://commitmenttoequity.org/datacenter/>). Adding other potential effects of aid on household income in general, the 24 percent estimate obtained for a reasonably well-governed country does not look unreasonable.

From a careful look at Fig. 1-4, the following observations emerge:

1. There is an inverse relationship between w_1 and s_1 .
2. For a given degree of poverty aversion, the differences in s_1 between the exogenous and endogenous monitoring regimes is larger as the gap in basic governance between the two countries is wider (β_1 is further away from β_2).
3. For a given level of basic governance, the gap in aid shares between a regime of zero monitoring and a regime of endogenous monitoring is increasing as the gap in pre-aid incomes of the poor is getting smaller (Proposition 7).
4. When poverty aversion is stronger ($\varepsilon = 3$), the differences in s_1 between the exogenous and endogenous monitoring regimes are extremely small, and they are all the smaller as β_1 gets closer to β_2 . The same holds true for the post-aid income of the poor whether in country 1 or country 2 (as shown for a unique value of β_1). The implication is that, as ε becomes sufficiently large, the endogenization of monitoring ceases to have a perceptible impact on aid shares. The rationale is as follows. A strongly poverty-averse donor is not much willing to accept a fall in the aid share of the poorer, worse-governed country, even while he resorts to strict monitoring in order to control aid misuse.
5. When poverty aversion is reasonably strong ($\varepsilon = 3$), the curves depicting s_1 as a function of w_1 intersect, meaning that s_1 under endogenous monitoring, from being higher than under exogenous monitoring, becomes smaller once w_1 is low enough (Proposition 5). The intersection points correspond to the critical values of w_1 at which s_1 is not affected by the switch of monitoring regime.
6. When poverty aversion is reasonably strong ($\varepsilon = 3$), the curves depicting the post-aid income z_2 as a function of w_1 may intersect, meaning that z_2 under endogenous monitoring, from being higher than under exogenous monitoring, may become smaller once w_1 is low enough. Moreover, if the initial value of w_1 is low, z_2 may not increase as w_1 rises.
7. The post-aid income of the poor in both countries, z_1 or z_2 , is lifted as w_1 increases.
8. The donor's switch to endogenous monitoring causes z_1 to increase but z_2 to decrease over the whole range of w_1 values. However, the inter-regime differential dwindles markedly when ε rises and, when $\varepsilon = 3$ and w_1 is below 70-80 percent of w_2 , it becomes almost imperceptible and the relationship between endogenous and exogenous monitoring may even be inverted as far as country 2 is concerned (Proposition 6).

7 Conclusion and policy implications

How to keep fragile countries eligible for aid otherwise than through the ex post introduction of ad hoc 'sweeteners' to aid allocation formal rules is a question facing all aid-givers who are poverty-averse and at the same time preoccupied by inefficiencies in aid use. The approach advocated in this paper pursues the well-established idea of a trade-off between needs and governance, yet at the same time it aims at enhancing the chances of doubly unfortunate countries -those which are relatively poor and ill-governed- to get access to aid, or to see their benefits from aid increase. This approach is actually well known among practitioners: instead of considering governance in poor countries as given, and allocate aid accordingly, the donor decides to exert influence on local aid-related governance through costly disciplining instruments based on monitoring and punishment. In the new setup, conventional concepts of aid effectiveness become misleading. The reason is that the level of aid embezzlement or misuse in the recipient countries is determined not only by their basic or overall governance, which is reasonably assumed to lie beyond external control, but also by the external discipline optimally chosen and imposed by the donor regarding aid use.

The adoption of a modified need-governance framework that embeds endogenous monitoring leads to results that significantly differ from those reached by the conventional approach in which monitoring is either absent or exogenously given. In this final section, attention is drawn to four key policy implications of the new approach proposed.

First, country aid shares are a dubious indicator of the benefits which a particular country earns through aid transfers. Since the donor has two instruments to pursue its objective of alleviating poverty -aid shares and the extent of tailor-made monitoring applied to each recipient country-, it will always be the case that, even when its aid share diminishes, the worse-governed country will gain from the shift from exogenous to endogenous monitoring. That is, the wellbeing of its poor will increase and, moreover, this increase will always be larger than the improvement of the poor's lot in the other, better-governed country. An immediate consequence is that aid-givers appealed by the idea of optimizing monitoring ought to stop conceiving their decisions in terms of country shares. Instead, because they would jointly determine the optimal level of monitoring and the country share for each and every recipient country, preferential treatment of a category of countries is not necessarily reflected

in their relative access to aid. The proper measure is the change in the standard of living of their poor as a result of both aid and monitoring efforts.

Second, the donor's preference for poverty reduction plays a key role in determining whether the switch to endogenous monitoring will perceptibly affect aid shares. A strong poverty aversion will thus make for small or even insignificant effects. This outcome, it bears stressing, does not come from a low weight attached to aid effectiveness considerations by the donor. On the contrary, a donor strongly bent on reducing poverty, and hence on lifting the incomes of the poor in the poorer and worse-governed country, wants to tighten external discipline when such a possibility is being opened. Where the difference lies between the behaviors of a strongly and a weakly poverty averse donor lies elsewhere: because the former derives a comparatively high marginal utility from an improvement in the poor's income, he is reluctant to (significantly) lower the aid share of the poorer country even while aid use is monitored (much) more strictly in that country. It would therefore be wrong to think that strong poverty aversion makes endogenous monitoring less appealing. This would be so only if the donor limits his interest in the amount of aid accruing to the poorer country, regardless of what portion of it reaches the poor, which is an obviously odd preference profile when monitoring is available.

Third, in the case where endogenous monitoring greatly affects outcomes, a distinct possibility is that the poor in the better-governed country will suffer a fall in their post-aid income as a result of the shift of monitoring regime. A conflict of interest would then arise between the poor in the two recipient countries, which does not imply, however, that the poor in the better-governed country would lose interest in the aid transfer. As a matter of fact, they always benefit from such a transfer in absolute terms. As for the elites, whichever the country to which they belong, they always lose from the regime shift. On the one hand, the amount of 'operable aid' (net of the monitoring cost) is smaller after governance has been endogenized and, on the other hand, their own take of the aid proceeds is getting reduced when monitoring is optimized.

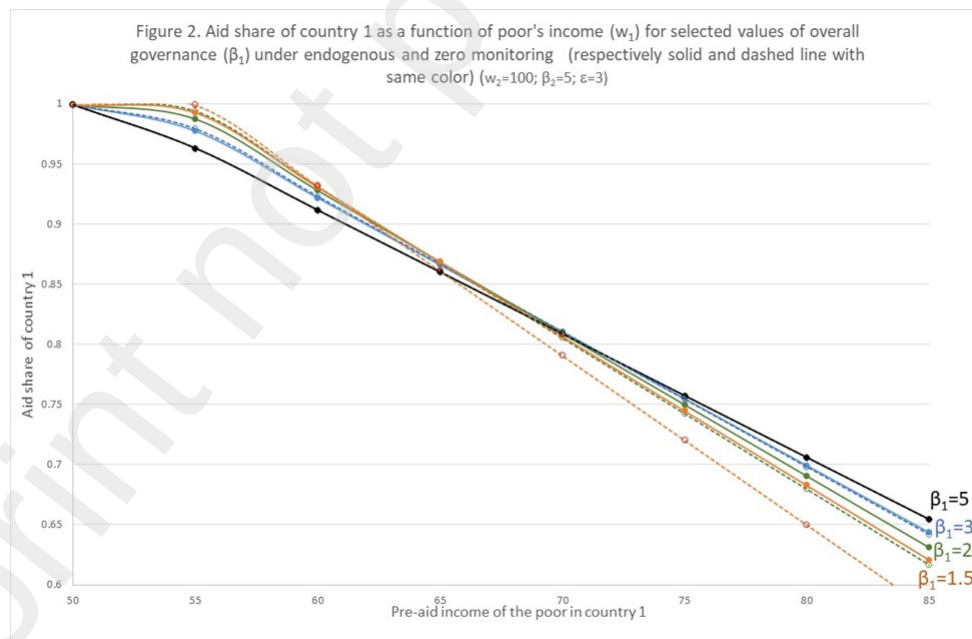
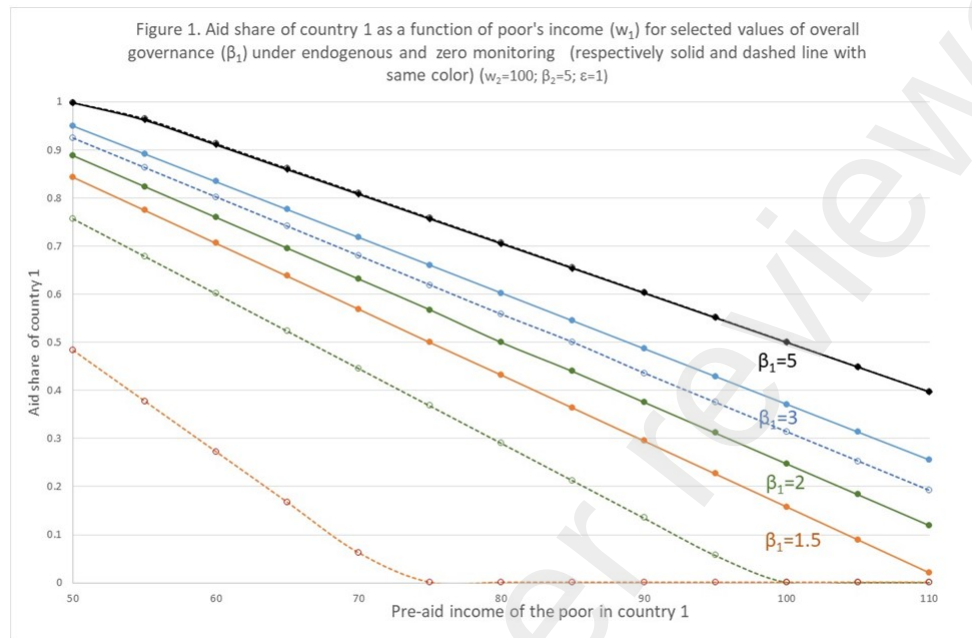
Fourth, the initial values of the inter-country gaps regarding key parameters of the model may determine the intensity, or even the direction of the effects at stake. Moreover, as witnessed by the varying slopes of the curves displayed in the figures, interaction effects between these parameters are typically at work. For example, for a given ratio of pre-aid per

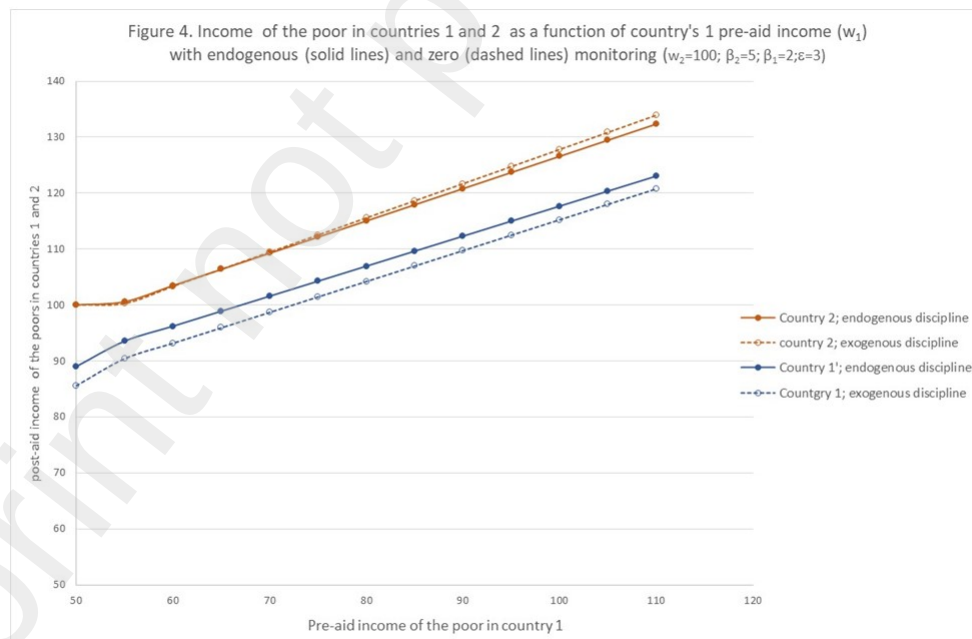
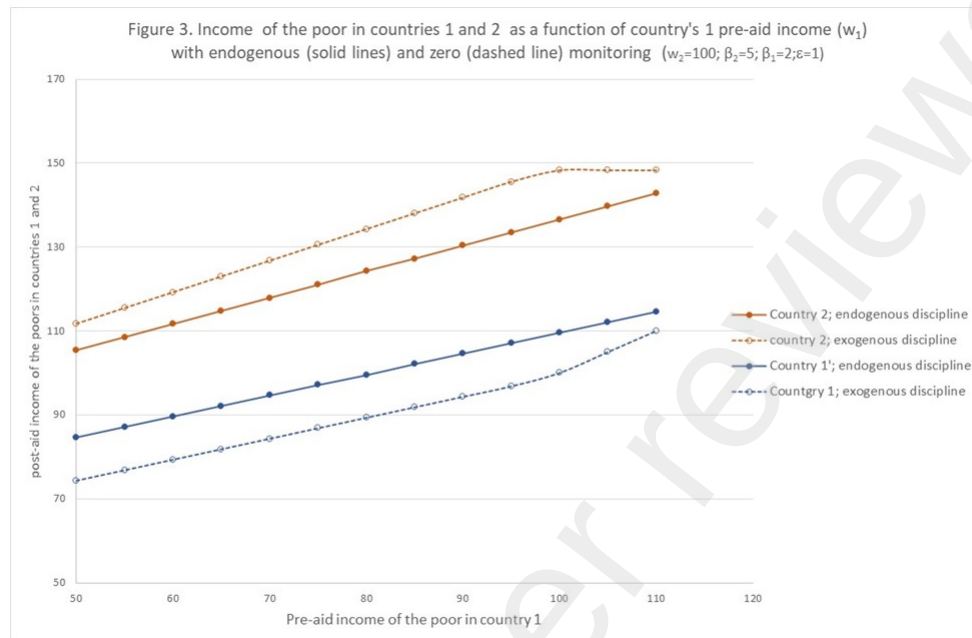
capita incomes of the poor, a wide (narrow) gap between basic governance levels will cause a large (small) differential between the country aid shares received under endogenous monitoring relative to those received under exogenous monitoring. There is an interaction between these two country parameters, however: the lower (higher) the pre-aid income in the worse-governed country compared to the pre-aid income in the better-governed country, the narrower (larger) this differential. In our second example, the interaction is between a country parameter and a donor's parameter. Thus, the fall of the post-aid per capita income of the poor in the better-governed country when the donor switches from exogenous to endogenous monitoring (see the third point above) can only occur if not only the donor's poverty aversion is strong enough but also the ratio of pre-aid incomes is low enough. The rationale behind this interaction is immediately intuitive.

The rigorous approach proposed in this paper clearly leads to results at odds with, and less trivial than those derived from standard PBA (performance-based) aid allocation rules. It also shows the important role of the donor's poverty aversion, a dimension of the problem which has been largely and strangely ignored by the literature. Considerable operational challenges are involved, among which the necessity to address serious measurement issues. In particular, there is a need to (i) define a proper way to assess the standard of living of poor people both before and after an aid intervention; (ii) to design a satisfactory method of estimating the degree of embezzlement or misuse of public resources in general (in order to assess basic or self-governance), or to translate existing indicators of governance quality into the shares of these resources that are appropriated by the local elite; and, last but not least, (iii) to estimate the monitoring cost function, implying the need to gain a more or less precise idea about the relationship between monitoring effort and improvement in the fraction of transfers that ends up with the poor themselves. The latter requires that the technology of supervision is specified. Finally, albeit not a measurement issue, the degree of poverty aversion of the donor is a crucial parameter, which needs to be made explicit at the time of the aid allocation-cum-monitoring decision.

Our hope is to have convinced the reader that tackling these difficult operational issues is a worthwhile task. As a matter of fact, the problem of fragile countries is going to stay with us for any foreseeable future, and adopting a systematic procedure that involves country-specific monitoring is better than relying on 'ad hoc' sweeteners to preserve aid eligibility of these

countries at all costs (or without adequate knowledge of the costs).





Appendix A: Proof of Proposition 5

Proof. Bearing in mind (3.8), the difference Δs_1 in the optimal shares of aid received by country 1 under the regimes of endogenous and exogenous monitoring can be written as follows:

$$\Delta s_1 = \frac{1}{D}(E_1^*/E_2^*)^{1/\varepsilon}(w_2/T + E_2^*) - \frac{1}{D^0}(E_1^0/E_2^0)^{1/\varepsilon}(w_2/T + E_2^0) - (w_1/T) \left(\frac{1}{D} - \frac{1}{D^0} \right),$$

where $D = E_2^*(E_1^*/E_2^* + (E_1^*/E_2^*)^{1/\varepsilon})$, and symmetrically for D^0 using E_1^0 and E_2^0 . If poverty aversion is less than or equal to unity, it is the case that $\frac{1}{D}(E_1^*/E_2^*)^{1/\varepsilon} < \frac{1}{D^0}(E_1^0/E_2^0)^{1/\varepsilon}$. This is because the function $g(x) = x^{1/\varepsilon}/(x + x^{1/\varepsilon})$ is decreasing when $\varepsilon \geq 1$ and $E_2^* > E_2^0$. Then, the preceding expression may be rewritten as:

$$\Delta s_1 = A(w_2/T) + B - C(w_1/T)$$

with:

$$A = \frac{1}{E_2}g[(E_1^*/E_2^*)] - \frac{1}{E_2^0}g[(E_1^0/E_2^0)], \quad B = g[(E_1^*/E_2^*)] - g[(E_1^0/E_2^0)], \quad C = \frac{1}{D} - \frac{1}{D^0}$$

It is easily checked that the A, B and C factors are all negative. This implies that if poverty aversion is above unity, Δs_1 is positive when w_1/T is above some positive threshold, $(A/C)w_2 + BT/C$, increasing in w_2 , and is negative below it. As this property is valid for $\varepsilon = 1$ it is also valid, by continuity, in some interval of ε below unity. There is uncertainty about the sign of Δs_1 for still lower values of poverty aversion. QED

□

Appendix A: Proof of Proposition 6

Proof. Denoting $z_i = w_i + s_i T E_i$ the post-aid income of country i , it comes from Proposition 3 - or (4.1) - and the first-order condition on aid shares (3.6) that:

$$\frac{z_1^*}{z_1^0} > \frac{z_2^*}{z_2^0}$$

where z_i^0 and z_i^* are respectively the optimal post-aid income under exogenous and endogenous monitoring. Clearly, the income of at least one country must increase with the endogenization of monitoring through the optimizing of the latter. From the preceding inequality, this must be the case of country 1, but not necessarily of the other country if its aid share declines - i.e. that of country 1 increases - and overweighs the gain arising from higher aid effectiveness.

To show that z_2^* may be smaller than z_2^0 is a bit intricate. Making use of (3.8), the optimal post-aid income of country 2 for given monitoring-adjusted aid effectiveness E_1 and E_2 is:

$$z_2 = w_2 + \frac{TE_1 - (E_1/E_2)^{1/\varepsilon} w_2 + w_1}{[E_1/E_2 + (E_1/E_2)^{1/\varepsilon}]}$$

The change in that post-aid income when switching from exogenous uniform to endogenous monitoring can be expressed as:

$$\Delta z_2 = -A'w_2 + B'T + C'w_1$$

where A' , B' , C' are given by:

$$A' = g(E_1^*/E_2^*) - g(E_1^0/E_2^0); B' = E_2^*h(E_1^*/E_2^*) - E_2^0h(E_1^0/E_2^0); \\ C' = k(E_1^*/E_2^*) - k(E_1^0/E_2^0)$$

□

with $h(x) = x/(x + x^{\frac{1}{\varepsilon}})$ and $k(x) = 1/(x + x^{\frac{1}{\varepsilon}})$. Assuming $\varepsilon \geq 1$, it is already known that $g(\cdot)$ is decreasing. In addition it can be seen that $h(\cdot)$ is increasing and $k(\cdot)$ is decreasing. The coefficients A' and C' are thus negative whereas B' is positive. It follows that Δz_2 is negative iff:

$$w_1 > \frac{A'}{C'} w_2 - \frac{B'T}{C'} = Y^1$$

In addition, the aid share of country 2 must be initially positive, which requires:

$$w_1 > (E_1^0/E_2^0)^{1/\varepsilon} w_2 - TE_1^0 = Y^2$$

Finally, it can be shown that there is a ceiling positive values of w_2 below which $Y^2 < Y^1$. Let $\bar{w}_2$ be that value, which is function only of the monitoring-adjusted aid effectiveness in the two countries under exogenous and endogenous monitoring. Then $w_2 < \bar{w}_2$ and $w_1 > Y^1$ guarantee that the post-aid income of the poor in the better-governed country is positive with exogenous monitoring and declines when monitoring is made endogenous. QED

Appendix C: Proof of Proposition 7

Proof. From (3.8), the derivatives of the optimal aid share of country 1 with respect to the pre-aid per capita incomes of the poor in the two countries are given by:

$$\frac{\partial s_1}{\partial w_1} = -\frac{1}{E_1^* + (E_1^*/E_2^*)^{1/\varepsilon} E_2^*}, \quad \frac{\partial s_1}{\partial w_2} = \frac{1}{E_1^* (E_1^*/E_2^*)^{-1/\varepsilon} + E_2^*}$$

Since $E_i^* > E_i^0$ and $E_1^*/E_2^* > E_1^0/E_2^0$, the denominator of the first expression is higher under optimal monitoring. In absolute terms, the value of $\frac{\partial s_1}{\partial w_1}$ is therefore smaller. Things are more intricate for the second expression. The denominator, which can be written as $E_2^*[(E_1^*/E_2^*)^{1-1/\varepsilon} + 1]$, is clearly larger under optimal monitoring if $\varepsilon \geq 1$. In this case, $\partial s_1/\partial w_2$ is again smaller compared to the alternative regime of uniform monitoring. The possibility nevertheless exists that a sufficiently low value of ε causes the denominator to be smaller, rather than higher, under optimal monitoring. It may then be noted that, when ε gets close to zero, the denominator tends towards infinity - i.e. $E_1/E_2 < 1$ - but it does so faster under exogenous monitoring since $E_1^0/E_2^0 < E_1^*/E_2^*$. By continuity, there exists a value of poverty aversion within the range of $[0, 1[$ below which the denominator of $\partial s_1/\partial w_2$ is smaller under optimal monitoring. It is therefore the case that $\partial s_1/\partial w_2$ is smaller under

optimal than under exogenous monitoring for all ε above the same threshold. QED □

Appendix D: Proof of Lemma 1 and Proposition 8

Proof. As shown in the proof of Proposition 1, the direction of change in the aid share of the worse-governed country when its basic governance has improved is given by the sign of $(1 - \mu_1 \varepsilon)$, where μ_1 is the net share of the aid amount in the post-aid income of its poor. Based on Proposition 1, denote ε^0 and ε^* the critical values of poverty aversion at which the aid share of the worse-governed country shifts from being increasing to being decreasing under the exogenous (uniform) and the endogenous monitoring regimes, respectively. Set $\varepsilon = \varepsilon^0$ and denote by μ_1^0 the corresponding value of μ_1 under exogenous monitoring, so that: $\mu_1^0 = s_1^0 T E_1^0 / z_1^0$. Keeping $\varepsilon = \varepsilon^0$, we examine the case of optimal country-specific monitoring. It was seen earlier that all parameters being the same, the optimal post-aid income of the poor in country 1 is then higher: $z_1^* > z_1^0$. As a result, the share of aid in the post-aid income of the poor in that country is such that $\mu_1^* > \mu_1^0$. For $\varepsilon = \varepsilon^0$, the following relationship consequently holds:

$$1 - \mu_1^* \varepsilon^0 < 1 - \mu_1^0 \varepsilon^0 = 0$$

In turn, this implies that $\varepsilon^* < \varepsilon^0$, and $\partial s_1 / \partial \beta_1$ is negative in the interval $[\varepsilon^*, \varepsilon^0]$ under endogenous monitoring while being positive under exogenous monitoring. QED □

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